JOHAAN TOMY KALLANY

PhD Applicant — Data-Driven Mathematical Modeling

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RESEARCH PROFILE

Graduate of the **Joint European MSc MathMods (Mathematical Modelling in Engineering)**, with academic training in ordinary and partial differential equations, numerical analysis and scientific computing, control theory, stochastic modelling and statistical learning. During my research internship, I developed numerical particle-dynamics solvers and combined Lagrangian particle statistics with Eulerian particle-density fields and carrier-flow diagnostics to quantify how flow structure, particle inertia and interparticle collisions shape collective organisation in prescribed carrier flows and wall-bounded turbulence. I aim to extend this mathematical and computational foundation towards **reinforcement learning** and **optimal control** of differential-equation governed systems under uncertainty.

RESEARCH EXPERIENCE

Research Intern — Inria Sophia Antipolis, France
APR. 2026 – SEPT. 2026

Team CALISTO · Supervisors: Dr Christophe Henry and Dr Laetitia Giraldi · Thesis Grade: 17/20

Thesis: ***Collective Dynamics of Colliding Inertial Particles in Prescribed Carrier Flows***

Research Output: First-author manuscript in preparation with L. Giraldi and C. Henry (Available at kallanyjohaan.github.io)

- **Developed and verified particle-dynamics solvers in Python**: a force-free event-driven reference and a kick–drift integrator coupling Stokes-drag ODEs with chronologically resolved collisions.
- **Reconstructed time-resolved wall-parallel velocity fields** from turbulent-channel DNS using validated space–time interpolation; computed flow gradients via FFT-based differentiation and near-wall streak scales from velocity autocorrelations.
- **Quantified collision-induced redistribution** through matched collision-on/off simulations, flow-conditioned statistics, coarse-grained number fluctuations, two-point density correlations and DBSCAN clustering.
- **Developed pair- and collision-event-conditioned statistics** for anisotropy, collision-angle selection and recurrence; assessed statistical uncertainty using moving-block bootstrap confidence bands and pair-cluster-robust standard errors.
- **Developed and implemented a reduced continuum model for particle-density evolution** in prescribed carrier flows via particle-response-time expansion and stochastic closure of unresolved collision histories through collision-localised random kicks; currently extending the framework to near-wall turbulence.

EDUCATION

Joint MSc in Mathematical Modelling in Engineering (MathMods)
SEPT. 2024 – SEPT. 2026

Université Côte d'Azur (France) · University of Hamburg (Germany) · University of L'Aquila (Italy) · mathmods.eu

Semester IV — Master's Thesis at Inria Sophia Antipolis — **APR. 2026 – SEPT. 2026**

Semester III — Université Côte d'Azur, France — **SEPT. 2025 – MAR. 2026**

Specialisation in **Stochastic Modelling and Statistical Learning**; selected cohort **rankings**: *Stochastic Calculus & Applications* (**2/15**), *Fundamentals of Machine Learning* (**2/20**), and *Statistical Learning from and for Graphs & Geometric Statistics* (**2/10**).

- **Stochastic Calculus, Diffusions & Control:** Brownian motion; Itô calculus and SDEs; Euler–Maruyama discretisation; infinitesimal generators; stochastic optimal control, Hamilton–Jacobi–Bellman equations and mean-field models.
- **Optimisation, Machine Learning & Geometric Statistics:** convex and large-scale optimisation; kernel methods and neural networks; variational inference; stochastic block models and graph learning; geometric statistics on manifolds.

Semester II — University of Hamburg, Germany — **APR. 2025 – AUG. 2025**

Received the **highest possible German grade (1.0)** in *Numerical Treatment of Ordinary Differential Equations, Probability-Theory* and the *Modelling Camp* research project.

- **Numerical Analysis & Machine Learning:** stiff and multistep integrators, DAEs; finite difference and finite-volume schemes for conservation laws; Grap-based learning, SVMs, PCA/SVD, Johnson–Lindenstrauss embeddings.

Semester I — University of L'Aquila, Italy — **SEPT. 2024 – FEB. 2025**

Awarded grade **30/30 cum laude in both** *Applied Partial Differential Equations* and *Control Systems,* and **29/30** in *Mathematical Modelling of Continuum Media.*

- **PDE Modelling & Control Theory:** Well-posedness of heat, wave and Laplace equations; maximum principles and Green's functions; incompressible Navier–Stokes and Stokes equations; controllability, Lyapunov stability and bifurcation theory.

BSc — Economics, Mathematics & Statistics

JUN. 2019 – MAY 2022

CHRIST (Deemed to be University), Bengaluru, India

CGPA 9.7/10; coursework in probability and stochastic processes, mathematical modelling, time-series forecasting and statistical computing in R and Python.

SELECTED COMPUTATIONAL RESEARCH PROJECTS

Hyperparameter Optimisation with Approximate Gradient (HOAG)

JAN. 2026 – FEB. 2026

Fundamentals of Machine Learning project · Université Côte d'Azur

- Analysed **bilevel hyperparameter optimisation** via implicit differentiation, including the $O(\varepsilon_k)$ hypergradient error and convergence to gradient-mapping stationarity under summable inner tolerances (Pedregosa, ICML 2016); derived a heuristic adaptive step-size rule and a strong-convexity stopping criterion.
- **Implemented HOAG in Python** for ℓ_2-regularised logistic regression and RBF kernel ridge regression using L-BFGS-B and matrix-free conjugate gradients, respectively; evaluated the method on high-dimensional sparse datasets.
- **Compared HOAG** against grid search, random search, Bayesian optimisation with a Gaussian-process surrogate, and iterative differentiation: the GP surrogate method reached the lowest suboptimality per unit runtime, while HOAG depended critically on a hand-tuned tolerance schedule.

Dynamic Micro–Macroscopic Hybrid Model for Efficient Highway Traffic Simulation

JUN. 2025 – SEPT. 2025

Modelling Camp research project · Department of Mathematics · University of Hamburg

- **Implemented a dynamic multi-fidelity micro–macro traffic framework** based on Imai et al. (2024), coupling the macroscopic Cell Transmission Model with the vehicle-level Intelligent Driver Model through adaptive resolution switching.
- **Developed and verified a Godunov finite-volume solver f**or the LWR conservation law against exact solutions, validating characteristic propagation, shock and rarefaction dynamics, and mass conservation.
- **Designed mass-preserving micro–macro interfaces and a fidelity-switching policy:** vehicle reservoirs with residual tracking; density-based hysteresis, dwell-time and coefficient-of-variation triggers confining microscopic resolution to congested regions; conservation verified across traffic regimes.

INDUSTRY EXPERIENCE

Associate Analyst — Deloitte Consulting (U.S.–India Office)

AUG. 2022 – JUN. 2024

Bengaluru, India

- Developed and maintained automated end-to-end regression tests for ERP data workflows, validating data ingestion and transformation to improve software reliability and reproducibility.
- Led a six-member team through monthly regression-test cycles, streamlining test execution, defect tracking and reporting to reduce overall cycle time.

RESEARCH & TECHNICAL SKILLS

- **Programming & Scientific Computing:** Python (NumPy, SciPy, PyTorch, scikit-learn), C++ (basic); Git, Linux, HDF5; Numba-accelerated physics-based particle simulators; solver verification against exact solutions and conservation checks.
- **Optimal Control & Dynamic Programming:** controllability and stabilisation of linear and nonlinear systems; dynamic programming principle, HJB equations; stochastic control of diffusions; mean-field models.
- **Mathematical Modelling:** Reduced continuum models from microscopic dynamics; stochastic closure of unresolved collision histories; hybrid continuous–discrete systems; conservative multi-fidelity PDE–ODE coupling.
- **Optimisation & Machine Learning**: bilevel optimisation (implicit differentiation, error bounds under inexact inner solves); L-BFGS-B, matrix-free CG; Bayesian optimisation (GP surrogates); neural-network training (PyTorch); DBSCAN clustering.
- **Numerical Analysis (ODEs, SDEs & PDEs):** stiff/IMEX integrators; Euler–Maruyama; Godunov schemes, CFL timestep control; operator splitting; event-driven simulation; FFT-based spectral differentiation; stability and convergence analysis.

AWARDS & SCHOLARSHIPS

Machine Learning + Simulation Thematic Semester — Inria (2026) · Selected and funded for master's thesis research with the CALISTO team at Inria Sophia Antipolis.

MAECI Scholarship (2024–25) — Italian Ministry of Foreign Affairs · Merit-based funding for tuition and living costs.

University Merit Scholarship (2020–21 & 2021–22) — awarded twice for outstanding academic performance at CHRIST (Deemed to be University).

REFERENCES

Prof. François Delarue, Université Côte d'Azur (Stochastic Analysis and Control) ·Dr. C. Henry & Dr. L. Giraldi, Inria Sophia Antipolis